

Tyre Request Management

Enterprise Custom ServiceNow Application Documentation

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EXECUTIVE SUMMARY

The **Tyre Request Management** application replaces manual tracking with a centralized, role-based workflow inside ServiceNow. Built with security and operational efficiency at its core, it leverages custom data models, granular Access Control Lists (ACLs), automated server-side business logic, and deployment-ready Update Sets to deliver a robust request fulfillment engine.



CORE OBJECTIVES

Centralized Administration

Single source of truth for all organizational tyre requests across departments and fleets.

Role-Based Security

Strict record and field-level access control via custom ACLs and application roles.

Automated Processing

Server-side logic via Business Rules to enforce strict data validation and auto-routing.

Seamless Deployment

Fully packaged into ServiceNow Update Sets for smooth migration across DEV, TEST, and PROD.



DATA ARCHITECTURE & SCHEMA

Custom Table: `u_tyre_request`

Field Label	Element Name	Data Type	Description / Notes
Request Number	<code>u_number</code>	Auto-Number	Unique identifier (e.g., TYR0001001)
Requested By	<code>u_requested_by</code>	Reference	Points to User record (<code>sys_user</code>)
Tyre Type	<code>u_tyre_type</code>	Choice	All-Terrain, Performance, Winter, Commercial
Tyre Size	<code>u_tyre_size</code>	String	Standard dimensions (e.g., 225/65R17)
Quantity	<code>u_quantity</code>	Integer	Number of units requested
Assignment Group	<code>u_assignment_group</code>	Reference	Points to Group table (<code>sys_user_group</code>)
Assigned To	<code>u_assigned_to</code>	Reference	Dependent on selected Assignment Group
State	<code>u_state</code>	Choice	Draft → Pending → In Progress → Completed
Priority	<code>u_priority</code>	Choice	Critical, High, Moderate, Low

ACCESS MANAGEMENT MATRIX

To maintain data integrity and satisfy audit compliance, security is enforced across four distinct user personas:

Persona / Role	Role Name	Create	Read	Write	Delete	Scope & Responsibilities
User	x_tyre_req.user	ALLOWED	OWN	OWN	DENIED	Creates requests; views and edits own records only
Agent	x_tyre_req.agent	ALLOWED	ALLOWED	ALLOWED	DENIED	Works assigned tickets; updates state & fulfillment notes
Manager	x_tyre_req.manager	ALLOWED	ALLOWED	ALLOWED	RESTRICTED	Oversees team requests, approves orders, cancels records
Admin	x_tyre_req.admin	ALLOWED	ALLOWED	ALLOWED	ALLOWED	Full admin rights, schema management, hard deletions

AUTOMATED BUSINESS LOGIC

Server-side **Business Rules** manage workflow validation and automation:

- **Data Validation (before insert / update)**: Validates that `u_quantity` is greater than `0` and standardizes text input prior to committing records.
- **Auto-Routing (before insert)**: Automatically populates `u_assignment_group` based on request location or vehicle parameters if left unassigned.
- **Deletion Guard (before delete)**: Prevents non-admin users from deleting active or completed requests to preserve audit compliance.

DEPLOYMENT & INSTALLATION GUIDE

Importing the Update Set:

1. Log in to the target instance as an **Administrator**.
2. Navigate to `System Update Sets` → `Retrieved Update Sets`.
3. Click **Import Update Set from XML** and upload the package.
4. Preview the Update Set, resolve any dependencies/conflicts, and click **Commit Update Set**.

FUTURE ENHANCEMENTS ROADMAP

Service Portal & MRVS

Expose ordering via Employee Center with Multi-Row Variable Sets for bulk requests.

Flow Designer & Approvals

Implement automated multi-tier approval workflows and manager sign-offs.

